

# SPORTS DATA SCIENCE

Understanding Player Load, Training Science  
& Performance Optimization



Sport Lisboa e Benfica · 2020

## WHAT IS SPORTS SCIENCE?

The study of human physiology, psychology, anatomy, biomechanics, biochemistry, and biokinetics to understand how the human body works during exercise.

### KEY GOAL

*Understand a player's LOAD — the sport and non-sport burden applied as a stimulus to the human body.*

1

#### External Load

What did a player do?

2

#### Internal Load

How much did it cost them?

# UNDERSTANDING LEADS TO INTERVENTIONS

INTERVENTION: Methods or programs to ensure athletes are in an optimal zone of fitness.

## Example Interventions could be:

01

### Training Adjustment

Adjusting the number of minutes of exercise as well as the volume and/or intensity — across training sessions, matches, and/or gym work.

02

### Recovery Methods

Implementing targeted recovery strategies to manage fatigue and restore athletes to peak performance between sessions.

## EXTERNAL & INTERNAL LOAD FRAMEWORK

### ▲ INTERNAL

#### Objective · Internal

Heart Rate Variability · Heart Rate Load · Resting Heart Rate · Blood Lactate

+Understands adaptation to fitness

—Measuring is difficult and invasive

#### Subjective · Internal

Wellness Survey · Session RPE

+Reliable and easy to implement

—Requires educating the athlete

#### Objective · External

GPS · Velocity Based Training · Accelerometer · Session Duration · Daily Vertical Jump

+Great measures of adaptation to fitness

—Invasive, but generally accepted

#### Subjective · External

Coach's Assessment

+Measures delta between coach and player perception

—Requires educating the coach

### ▼ EXTERNAL

# PHYSIOLOGICAL VS. MECHANICAL LOAD & FATIGUE

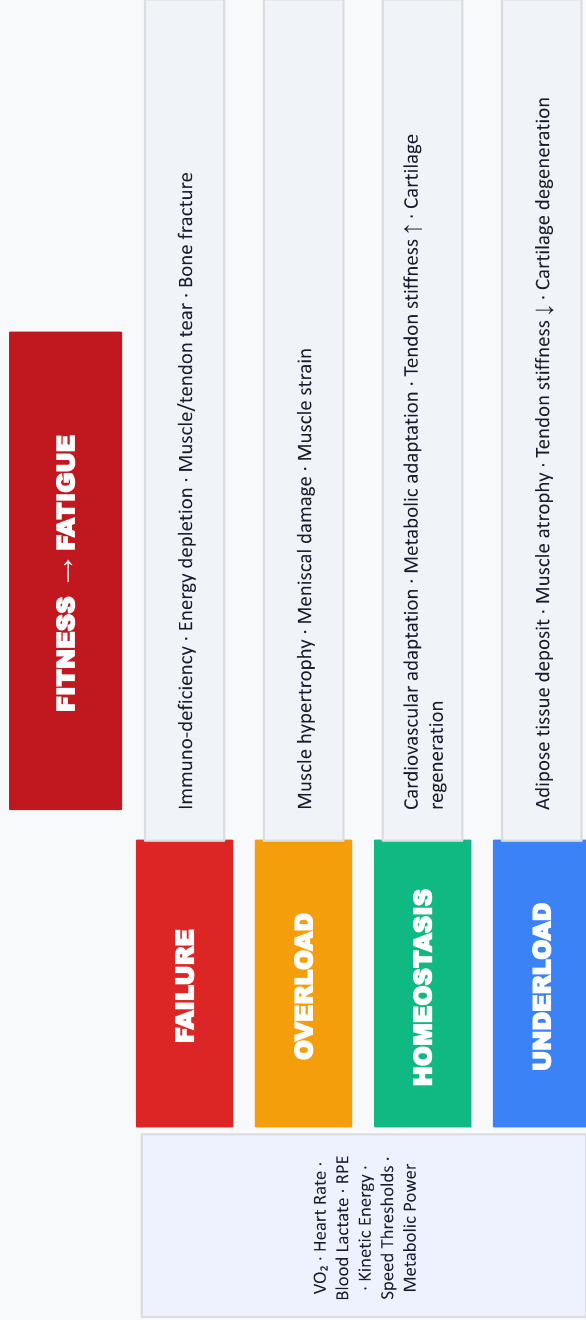
Athletes have two distinct reserves that are depleted by training and competition:

| Physiological Reserve                                            | Mechanical Reserve                                                 |
|------------------------------------------------------------------|--------------------------------------------------------------------|
| Cardiovascular & metabolic capacity                              | Musculo-skeletal integrity                                         |
| Depleted by aerobic / anaerobic demands                          | Depleted by impacts, sprints, jumps                                |
| Recovered through rest, sleep, nutrition                         | Recovered through structured rest days                             |
| Monitored via HR, lactate, HRV                                   | Monitored via GPS accelerometer load                               |
| Overload leads to overtraining syndrome                          | Overload leads to stress fractures & tears                         |
| External Load (What they DO)                                     | Internal Load (Cost to the body)                                   |
| Volume • Intensity • Distance • Contacts • Jumps • Accelerations | Heart Rate • RPE • Blood Lactate • Cortisol • HRV • Wellness Score |

# TRACKING TECHNOLOGY

|                                                                                                                                                                                                             |                                                                                                                                                                                                         |                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br><b>GPS Tracker</b><br>Receives signals from GPS & GLONASS satellites for dual-system tracking with high precision. | <br><b>Gyroscopes</b><br>Measure orientation of the athlete's body position during movement.                         | <br><b>Accelerometers</b><br>Measure impact forces across all planes of movement and direction changes.        |
| <br><b>Magnetometers</b><br>Measure direction like a digital compass embedded in the wearable pod.                     | <br><b>Microprocessor</b><br>Central processing unit inside the vest pod that crunches all sensor data in real time. | <br><b>Velocity-Based Training</b><br>Monitors bar speed and power output during strength sessions in the gym. |

# FITNESS-FATIGUE FRAMEWORK



## TRAINING EFFICIENCY INDEX (TEI)

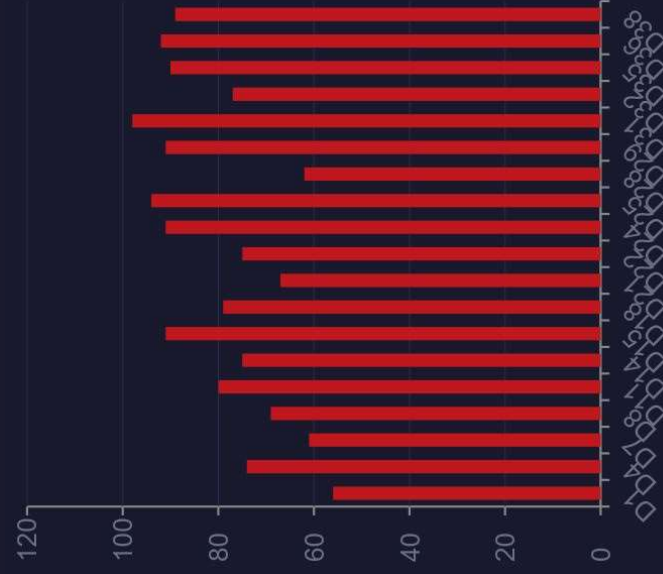
A log-linear relationship between External Load and Internal Load helps model how an athlete is reacting to a stimulus — to determine under-loading, over-loading, or sustained optimal loading.

**0.85**  
Default Slope

**0.81**  
Measured Slope

**0.93**  
R Squared

**82.55**  
Average TEI



# TRAINING PERIODIZATION

← 1 YEAR (MACROCYCLE) →

## MACROCYCLE

## MESOCYCLES

## MICROCYCLES

## INDIVIDUAL SESSIONS

How coaches structure training across different time horizons:

### Macrocycle

The full season plan (~1 year)

### Mesocycle

Multi-week training blocks (4–6 weeks)

### Microcycle

Week-by-week training structure

### Sessions

Individual training units

# RPE & MATCH-DAY TRAINING LOAD



01

Peak load occurs at MD-3, reflecting high-intensity tactical sessions mid-week.

02

Load tapers significantly on MD-2 and MD-1 to prepare players for match day.

03

RPE and session-RPE together track both intensity and total workload effectively.

# THE FUTURE: NON-INVASIVE TRACKING

Computer vision and optical tracking systems are enabling teams to capture GPS-equivalent data — without requiring players to wear any physical devices.

## Computer Vision



Stadium cameras track all 22 players simultaneously at high frame rates.

## AI Pose Estimation



Deep learning models estimate joint angles, speed, and workload in real time.

## Privacy-Friendly



No wearables required — reduces friction and ethical concerns around athlete data.

## Match & Training



Works seamlessly in both competitive matches and closed training sessions.

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## MOVING TO A MORE NON-INVASIVE METHOD

Modern tracking technology allows performance staff to collect high-resolution data without any physical contact with the athlete.

### Computer Vision & Optical Tracking

Cameras around the pitch track all players at 25 fps

No wearable device required

Provides x-y coordinates for every player and the ball

Used by Hawkeye, SciSports, Tracab, Second Spectrum

Enables Voronoi diagrams, pressure maps, and space metrics

### AI-Powered Video Analysis

Deep learning models detect player poses from broadcast footage

Estimates joint angles without motion capture suits

Scalable to any game footage worldwide

Used for biomechanical screening at scale

Reduces cost of lab-based analysis by ~90%

### Wearable + Optical Fusion

Combines GPS vest data with camera tracking

Cross-validates positional accuracy

Adds physiological context (HR, load) to spatial data

Standard in elite football, rugby, and basketball

Live dashboards streamed to coaches during training

*Key Advantage: Non-invasive methods reduce athlete burden, maintain data collection during competition, and allow retrospective analysis of archived footage — transforming how clubs scout, train, and rehabilitate players.*

# TRAINING DRILL OPTIMIZER

## Drill Calculator

The calculator allows a coach to choose a drill and receive the predicted external load per minute.

| Drill Inputs                                                                                                           |  | Predicted Output             |
|------------------------------------------------------------------------------------------------------------------------|--|------------------------------|
| Select Match Day<br>1                                                                                                  |  | Number of Minutes<br>15 min  |
| Select Athletes<br>Player 12, Player 17, Player 23 ...                                                                 |  | Drill Width<br>30 m          |
| Select Drill Category<br>CCO – Defesa em Bloco                                                                         |  | Drill Length<br>25 m         |
| Select Drill<br>Org. Defensiva Estrategia C Trans. Df Def                                                              |  | Predicted Load<br>7.4 AU/min |
| Select Format<br>4v4+GKs                                                                                               |  | Total Session Load<br>111 AU |
| Objective: SEI • Drill Duration — Uses historical GPS data to model load per drill type, format, and pitch dimensions. |  |                              |